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Summer internship report

Zied Manai

Hosting Enterprise:
SQOIN blockchain and cryptocurrency

SQOIN
Blockchain &
Cryptocurrency

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I - Acknowledgement:

The internship opportunity I had with SQOIN was a great chance for learning from professional experiences. Therefore, I consider myself as a very lucky individual as I was provided with an opportunity to be a part of it. I am also grateful for having a chance to meet so many wonderful people and professionals who led me though this internship period. Bearing in mind previous I am using this opportunity to express my deepest gratitude and special thanks to my supervisor and the CEO of SQOIN Mr. Bacem Bergaoui for always believing in me and for his precious guidance through this remarkable journey.
My huge thanks go to all the team of SQOIN for making for me a welcoming and lovely environment to work and to integrate.

II - Host enterprise:

Born from the union between experienced computer engineers and experienced academics, SQOIN is a Tunisian startup present on the international market of blockchain and cryptocurrency.
SQOIN operates mainly on four axes:

- Development and management of Blockchain, cryptocurrency and FinTech services projects.
- Technical support and customer follow-up (Assistance center, Incident management, Configuration management).
- Promotion and marketing of products designed and developed by SQOIN: the "Bostoji" cryptocurrency and the "Sqoin Exchange".
- Sqoin Academy: certified training courses on blockchain and cryptocurrency themes

SQOIN has completed and ensured the success of dozens of high added value IT projects on behalf of several European and American firms operating in the field of Blockchain and digital currencies. Among the successful clients, we cite:

- Host Master Node India
- Sage city United Kingdom
- Demyestify Canada

III - The bitcoin Blockchain:

1 - Introduction:

Bitcoin is a collection of concepts and technologies that form the basis of a digital money ecosystem. Units of currency called bitcoin are used to store and transmit value among participants in the bitcoin network. Bitcoin users communicate with each other using the bitcoin protocol primarily via the internet, although other transport networks can also be used. The bitcoin protocol stack, available as open source software, can be run on a wide range of computing devices, including laptops and smartphones, making the technology easily accessible.

Bitcoin was invented in 2008 with the publication of a paper titled “Bitcoin: A Peer-to-Peer Electronic Cash System,” written under the alias of Satoshi Nakamoto. Nakamoto combined several prior inventions such as b-money and HashCash to create a completely decentralized electronic cash system that does not rely on a central authority for currency issuance or settlement and validation of transactions. The key innovation was to use a distributed computation system (called a “Proof-of-Work” algorithm) to conduct a global “election” every 10 minutes, allowing the decentralized network to arrive at consensus about the state of transactions.

Users can transfer bitcoin over the network to do just about anything that can be done with conventional currencies, including buy and sell goods, send money to people or organizations, or extend credit. Bitcoin can be purchased, sold, and exchanged for other currencies at specialized currency exchanges. Bitcoin in a sense is the perfect form of money for the internet because it is fast, secure, and borderless.

2 - Keys, Addresses:

Public key cryptography:

Public key cryptography was invented in the 1970s and is a mathematical foundation for computer and information security. Since the invention of public key cryptography, several suitable mathematical functions, such as prime number exponentiation and elliptic curve multiplication, have been discovered. These mathematical functions are practically irreversible, meaning that they are easy to calculate in one direction and infeasible to calculate in the opposite direction. Based on these mathematical functions, cryptography enables the creation of digital secrets and unforgettable digital signatures. Bitcoin uses elliptic curve multiplication as the basis for its cryptography.

In blockchain, we use public key cryptography to create a key pair that controls access to wallets. The key pair consists of a private key and—derived from it—a unique public key. The public key is used to receive funds, and the private key is

used to sign transactions to spend the funds. There is a mathematical relationship between the public and the private key that allows the private key to be used to generate signatures on messages. This signature can be validated against the public key without revealing the private key.
When creating a transaction, the sender presents his public key and a signature (different each time, but created from the same private key) in a transaction. Through the presentation of the public key and signature, everyone in the blockchain network can verify and accept the transaction as valid, confirming that the transaction is valid

Private Keys:

A private key is simply a number, picked at random. Ownership and control over the private key is the root of user control in the blockchain. The private key is used to create signatures that are required to approve the transaction by proving ownership of funds used. The private key must remain secret at all times, because revealing it to third parties is equivalent to giving them control over the wallet secured by that key. The private key must also be backed up and protected from accidental loss, because if it's lost it cannot be recovered and the funds secured by it are forever lost, too.

Public Keys:

The public key is calculated from the private key using elliptic curve multiplication, which is irreversible: K = k * G, where k is the private key, G is a constant point called the generator point, and K is the resulting public key. The reverse operation, known as "finding the discrete logarithm"—calculating k if you know K—is as difficult as trying all possible values of k, i.e., a brute-force search. Before we demonstrate how to generate a public key from a private key, let's look at elliptic curve cryptography in a bit more detail.

Bitcoin Addresses:

A bitcoin address is a string of digits and characters that can be shared with anyone who wants to send you money. Addresses produced from public keys consist of a string of numbers and letters, beginning with the digit "1." Here's an example of a bitcoin address:

1J7mdg5rbQyUHENYdx39WVWK7fsLpEoXZy

The bitcoin address is what appears most commonly in a transaction as the "recipient" of the funds. If we compare a bitcoin transaction to a paper check, the bitcoin address is the beneficiary, which is what we write on the line after "Pay to the order of." On a paper check, that beneficiary can sometimes be the name of a bank account holder, but can also include corporations, institutions, or even cash. Because paper checks do not need to specify an account, but rather use an abstract name as the recipient of funds, they are very flexible payment

instruments. Bitcoin transactions use a similar abstraction, the bitcoin address, to make them very flexible.

3 - Wallets:

A common misconception about blockchain is that wallets contain coins. In fact, the wallet contains only keys. The “coins” are recorded in the blockchain network. Users control the coins on the network by signing transactions with the keys in their wallets. In a sense, a wallet is a keychain.
There are two primary types of wallets, distinguished by whether the keys they contain are related to each other or not. The first type is a nondeterministic wallet, where each key is independently generated from a random number. The keys are not related to each other. This type of wallet is also known as a JBOK wallet from the phrase “Just a Bunch Of Keys.” The second type of wallet is a deterministic wallet, where all the keys are derived from a single master key, known as the seed. All the keys in this type of wallet are related to each other and can be generated again if one has the original seed. There are a number of different key derivation methods used in deterministic wallets. The most commonly used derivation method uses a tree-like structure and is known as a hierarchical deterministic or HD wallet. Deterministic wallets are initialized from a seed. To make these easier to use, seeds are encoded as English words, also known as mnemonic code words.

Seeds and Mnemonic Codes (BIP-39):

HD wallets are a very powerful mechanism for managing many keys and addresses. They are even more useful if they are combined with a standardized way of creating seeds from a sequence of English words that are easy to transcribe, export, and import across wallets. This is known as a mnemonic and the standard is defined by BIP-39. Today, most bitcoin wallets (as well as wallets for other cryptocurrencies) use this standard and can import and export seeds for backup and recovery using interoperable mnemonics.
Let’s look at this from a practical perspective. Which of the following seeds is easier to transcribe, record on paper, read without error, export, and import into another wallet?

0C1E24E5917779D297E14D45F14E1A1A

army van defense carry jealous true garbage claim echo media make crunch

4 - Transactions:

Transactions are the most important part of the blockchain. Everything else in blockchain is designed to ensure that transactions can be created, propagated on the network, validated, and finally added to the global ledger of transactions. Transactions are data structures that encode the transfer of value between participants. Each transaction is a public entry in the blockchain.

In this chapter we will examine all the various forms of transactions, what they contain, how to create them, how they are verified, and how they become part of the permanent record of all transactions.

5 - The blockchain network:

Peer-to-Peer Network Architecture:

Bitcoin is structured as a peer-to-peer network architecture on top of the internet. The term peer-to-peer, or P2P, means that the computers that participate in the network are peers to each other, that they are all equal, that there are no “special” nodes, and that all nodes share the burden of providing network services. The network nodes interconnect in a mesh network with a “flat” topology. There is no server, no centralized service, and no hierarchy within the network. Nodes in a P2P network both provide and consume services at the same time with reciprocity acting as the incentive for participation. P2P networks are inherently resilient, decentralized, and open. A preeminent example of a P2P network architecture was the early internet itself, where nodes on the IP network were equal. Today’s internet architecture is more hierarchical, but the Internet Protocol still retains its flat-topology essence.

Beyond bitcoin, the largest and most successful application of P2P technologies is file sharing, with Napster as the pioneer and BitTorrent as the most recent evolution of the architecture. Bitcoin’s P2P network architecture is much more than a topology choice. Bitcoin is a P2P digital cash system by design, and the network architecture is both a reflection and a foundation of that core characteristic. Decentralization of control is a core design principle that can only be achieved and maintained by a flat, decentralized P2P consensus network. The term “bitcoin network” refers to the collection of nodes running the bitcoin P2P protocol. In addition to the bitcoin P2P protocol, there are other protocols such as Stratum that are used for mining and lightweight or mobile wallets. These additional protocols are provided by gateway routing servers that access the bitcoin network using the bitcoin P2P protocol and then extend that network to nodes running other

Node Types and Roles:

Although nodes in the bitcoin P2P network are equal, they may take on different roles depending on the functionality they are supporting. A bitcoin node is a collection of functions: routing, the blockchain database, mining, and wallet services. All nodes include the routing function to participate in the network and might include other functionality. All nodes validate and propagate transactions and blocks, and discover and maintain connections to peers. Some nodes, called full nodes, also maintain a complete and up-to-date copy of the blockchain. Full nodes can autonomously and authoritatively verify any transaction without external reference. Some nodes maintain only a subset of the blockchain and verify

transactions using a method called simplified payment verification, or SPV. These nodes are known as SPV nodes or lightweight nodes. Mining nodes compete to create new blocks by running specialized hardware to solve the Proof-of-Work algorithm. Some mining nodes are also full nodes, maintaining a full copy of the blockchain, while others are lightweight nodes participating in pool mining and depending on a pool server to maintain a full node. The mining function is shown in the full node as a black circle called “Miner” or the letter “M.”
In addition to the main node types on the bitcoin P2P protocol, there are servers and nodes running other protocols, such as specialized mining pool protocols and lightweight client-access protocols.

6 - Mining and consensus:

The word “mining” is somewhat misleading. By evoking the extraction of precious metals, it focuses our attention on the reward for mining, the new bitcoin created in each block. Although mining is incentivized by this reward, the primary purpose of mining is not the reward or the generation of new coins. If you view mining only as the process by which coins are created, you are mistaking the means (incentives) as the goal of the process. Mining is the mechanism that underpins the decentralized clearinghouse, by which transactions are validated and cleared. Mining is the invention that makes bitcoin special, a decentralized security mechanism that is the basis for P2P digital cash. Mining secures the bitcoin system and enables the emergence of network-wide consensus without a central authority. The reward of newly minted coins and transaction fees is an incentive scheme that aligns the actions of miners with the security of the network, while simultaneously implementing the monetary supply.
Miners validate new transactions and record them on the global ledger. A new block, containing transactions that occurred since the last block, is “mined” every 10 minutes on average, thereby adding those transactions to the blockchain. Transactions that become part of a block and added to the blockchain are considered “confirmed,” which allows the new owners of bitcoin to spend the bitcoin they received in those transactions.

IV - Wesharish coin:

1 - Introduction:

One of UNICEF's sustainable development goal is No poverty. UNICEF is working toward that goal by providing financial aid support for different communities around the word. Despite these efforts, the large number of middleman between the funder and the true beneficiaries ensures that a large amount of money is lost in transactions and not finding their way to the poor community. To solve this problem SQOIN come up with the idea of Wesharish coin. UNICEF funds are converted to Wesharish coins then these coins are sent to beneficiaries into their wallets on their mobile phones. If the benificairy doesn't have a smartphone he is provided with a paper wallet. Poor people can as consequence buy from local businesses using a digital currency helping small businesses to grow. Local traders can cash out their cumulated coins into real currency.

2 - Bitcoin core:

Bitcoin is an open source project and the source code is available under an open (MIT) license, free to download and use for any purpose. Open source means that bitcoin is developed by an open community of volunteers. At first, that community consisted of only Satoshi Nakamoto. By 2016, bitcoin's source code had more than 400 contributors with about a dozen developers working on the code almost full-time and several dozen more on a part-time basis.
Bitcoin Core's source code can be cloned from the source repository on GitHub. We can start the build process by generating a set of build scripts using the autogen.sh script. The autogen.sh script creates a set of automatic configuration scripts that will interrogate our system to discover the correct settings and ensure we have all the necessary libraries to compile the code. The most important of these is the configure script that offers a number of different options to customize the build process. The autogen.sh script creates a set of automatic configuration scripts that will interrogate our system to discover the correct settings and ensure we have all the necessary libraries to compile the code. The most important of these is the configure script that offers a number of different options to customize the build process. The Bitcoin Core client implements a JSON-RPC interface that can also be accessed using the command-line helper bitcoin-cli . The command line allows us to experiment interactively with the capabilities that are also available programmatically via the API. We can check the blockchain height, create transactions, explore blocks and many more.

3 - The forked coin:

Wesharish coin is forked from the bitcoin open source code. A part of the internship was to adapt this code to our specific need. To make the new blockchain independent of the bitcoin blockchain we generated a new genesis block other than Satushi's first block. In addition the private keys generation mechanism had to be changed in order to create new addresses different from bitcoin addresses to be used by our blockchain. Besides the default minimum difficulty configuration was too high for ordinary computers to handle in acceptable time, so we had to low it. Our need is to create a coin to be used for charity, so we had to set the transaction fees to 0. The code must also connect to our own DNS seeders. The default seeders of bitcoin were replaced by the addresses of our AWS nodes. The C++ code was complied and an executable script was generated. Now our blockchain is ready, and it needs to be deployed on different nodes in order to be tested.

4 - Mobile wallet:

The mobile wallet code was forked from the Breadwallet opensource code. A lot of work had to be done on the Mobile app in order to remove all the coins natively supported by Breadwallet and keep only the bitcoin wallet. We had to remove unnecessary functionalities such us trading that we don't need. Then we hand to change the code to generate new private keys different from those supported by bitcoin. We faced a lot of problems connection the app to the blockchain network. We passed a lot of time in the debugging step to finally solve the problem.

V - Solution deployment:

1 - AWS IaaS:

Amazon Web Services (AWS) is a subsidiary of Amazon providing on-demand cloud computing platforms and APIs to individuals, companies, and governments, on a metered pay-as-you-go basis. These cloud computing web services provide a variety of basic abstract technical infrastructure and distributed computing tools. One of these services is Amazon Elastic Compute Cloud (EC2), which allows users to have at their disposal a virtual cluster of computers, available all the time, through the Internet.
IaaS is a form of cloud computing that provides virtualized computing resources. We essentially use IaaS to replace physical resources, such as servers, with virtual resources hosted and managed by Amazon.
For the sake of deploying our blockchain we created 5 AWS IaaS nodes. The nodes were secured using TLS

2 - Docker:

Docker is a set of platform as a service (PaaS) products that use OS-level virtualization to deliver software in packages called containers. Containers are isolated from one another and bundle their own software, libraries and configuration files. They can communicate with each other through well-defined channels. All containers are run by a single operating system kernel and therefore use fewer resources than virtual machines. Docker is software that runs on Linux and Windows. It creates, manages and orchestrates containers. The software is developed in the open as part of the Moby open-source project on GitHub. Docker, Inc. is a company based out of San Francisco and is the overall maintainer of the open-source project. Docker, Inc. also offers commercial versions of Docker with support contracts etc.

3 - The solution:

The obtained executable script is deployed on the AWS. We faced a lot of the problem of missing libraries although the system is operational on our local machines. The problem is that the code was compiled on Ubuntu 20.04 LTS but AWS nodes were running on Ubuntu 18.04 LTS. The idea was to deploy the Wesharish core on a docker container. We specified the needed Ubuntu version and the needed dependencies in Dockerfiles and exported the port to be used by our blockchain. The images were build and executed on AWS node.
One node was reserved for the mining task. This node finds any pending transactions in the memory pool and generate a new block. Generated blocks can be seen by all the nodes of the network which means that they are synchronized. Transactions created by one node were mined by the mining node and the amount was added to the wallet of the reserving node. The system is fully operational.

VI - Conclusion:

Blockchain technology is an emerging technologies that allowed humanity for the first time in history to eliminate the need of a trusted third party. The idea of decentralized trust allowed the creation of innovative business models for solving many of humanities enormous problems. The promises of this technology are huge ranging from charity in the benefit of the isolated and the most valuable ones to application involving physical connected things. This potential made a lot of experts think that blockchain is the new internet. This internship was a good opportunity to get into the world of decentralization and to work on real application for the benefit of needy people. When this report was written wesharish project was selected among the top 100 project in the UNICEF competition and still waiting for the final result, making SQOIN the first Tunisian startup to reach this level in a call for action organized by UNICEF